

Final Exam in INF-2700

Database Systems

2011-12-09

dummy page

1 (40%)

Below are some database tables for a micro-blog web application. The example data show that blog 101 was commented by two blogs 102 and 105, which were in turn commented by blogs 106 and 110. A commenting blog gives also a rate to the blog it comments. For example, blog 110 comments blog 105 and rates blog 105 with 3 points.

- Writers

Wid	Name
7	cook
9	food_lover
20	vegi

- Blogs

Bid	Author	Content	Date
101	7	Here is a biff menu	08-12-2011
102	9	yum-yum	08-12-2011
105	20	biff is not healthy	09-12-2011
106	7	thanks for encouragement	09-12-2011
110	9	why not?	09-12-2011

- Comments

This	Other	Other_rate
102	101	10
105	101	0
106	102	9
110	105	3

The *primary keys* of the tables are in **bold** font.

Foreign keys:

- Blogs

- **Author**: references Wid of Writers

- Comments

- **This**: references Bid of Blogs

- **Other**: references Bid of Blogs

This blog is a comment on the **Other** blog and gives a rate on the **Other** blog.

Write queries to find the required information.

Queries 1–5 must be formulated in *both relational algebra and SQL*.

Queries 6–10 need only be formulated in *SQL*.

1. List the names of all writers.
2. Given the Wid of a writer, list the dates on which he/she wrote some blog(s).
3. Given the Bid of a blog (for example 101), list the contents and rating points of all its commenting blogs.

A commenting blog comments on another blog. For example, blog 110 comments on blog 105 with rating point 3.

For blog 101, the result is

Content	Rate
yum-yum	10
biff is not healthy	0

4. Bids of all original blogs.
An original blog does not comment on any other blog. Blog 101 is an original blog.
5. Given the Wid of a writer, list the names of all his/her fans.
 A is B 's fan if A has commented on B 's blog(s) and all A 's comments on B 's blogs are rated between 8 and 10. For example, cook is food_lover's fan and food_lover is also cook's fan.
6. Given the Bid of a blog, find the highest and the lowest rate on it.
7. Given the Bid of a blog, find its popularity.
The popularity of a blog is defined as the number of comments on it. For example, the popularity of blog 101 is 2.
8. Find blogs that have popularity greater than n .
9. List the bids and popularities of blogs in the descending order of their popularity.
10. List the names of all positive commenters.
A writer is a positive commenter if the average rate he/she gives to other blogs is 7 or above.

2 (20%)

We now consider the physical design of the **Comments** table. Assume that we decide to use *hash file organization* with hashing on attribute **Other**. In addition, we would like to have a *hash index* on attribute **This**.

Answer the following questions. To keep your answers focused, consider only static hashing.

1. How are **Comments** data organized on disk?
2. How to search data records on attribute **Other**?
3. What is the performance overhead of a search on **Other**?
4. How is the index on **This** organized?
5. How to search data records on attribute **This**?
6. What is the performance overhead of a search on **This**?

3 (20%)

A	B	C
1	x	t
1	y	u
2	z	u

1. What is *functional dependency* $P \rightarrow Q$ of a relation instance r ?
Given the above relation instance, check if the following functional dependencies are satisfied:
 - $A \rightarrow B$
 - $B \rightarrow C$
 - $AC \rightarrow B$
 - $AB \rightarrow B$
2. What do we mean by saying the following?
Relation schema R is in *Boyce-Codd Normal Form* (BCNF) with respect to a set of functional dependencies F .
3. What do we mean by saying the following?
Relation schema R is in *Third Normal Form* (3NF) with respect to a set of functional dependencies F .
4. Discuss the advantages and disadvantages of BCNF and 3NF.

4 (20%)

1. What is an *ACID transaction*?
2. What is a *log* for transaction processing?
How is a log organized?
What are the operations on a log?
What are the performance costs of these operations?
3. What is a *checkpoint* in a log?
4. What is *fuzzy checkpointing*? What problem does it address?